

Sets, sample spaces and events

Write every outcome as an element of a set. Events are then subsets, and set operations mean or, and, not.

THE QUESTION

What exactly do we give a probability to, and which rules must those numbers follow?

THE INTUITION

List every outcome: that set is the sample space S . An event is any subset of S , such as "even" = {2, 4, 6} on a die. Or, and, not become union, intersection, complement, and three axioms fix how probabilities combine.

KEY TAKEAWAY

Write each event as a set of outcomes. With equally likely outcomes, $P(E) = |E| / |S|$; the other rules follow from the axioms.

experiment → sample space S → event $E \subseteq S$ → or · and · not → three axioms → $P(E) = |E| / |S|$ → simulate to check

1 A set holds distinct elements, in no order

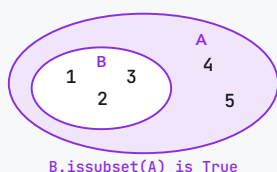
list [3, 1, 2, 3, 1]

3 1 2 3 1

repeats marked

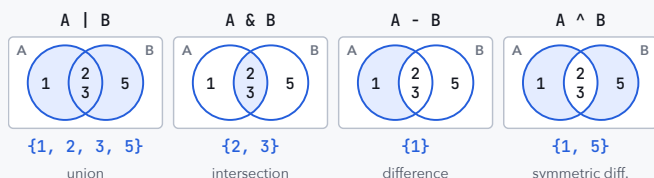
→ set(...) or {3, 1, 2, 3, 1}

1 2 3 3 elements, no order



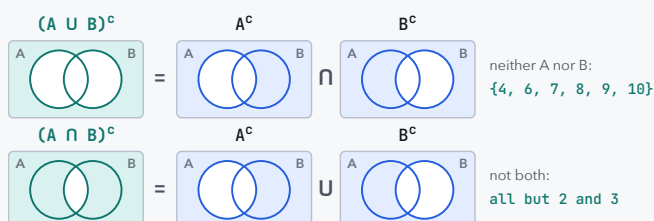
{3, 1, 2, 3, 1} and set([3, 1, 2, 3, 1]) both give {1, 2, 3}, with no index ($A[0]$ fails). $B.issubset(A)$, or $B \leq A$, asks if every element of B is in A ; $B < A$ is a proper subset ($B \neq A$). The empty set is a subset of every set. Traps: {} is an empty dict, so write set(); a set cannot hold a set ("unhashable type"), so use frozenset.

2 Union, intersection, difference



With $A = \{1, 2, 3\}$ and $B = \{2, 3, 5\}$: union, in A or B ; intersection, in both; difference, in A but not B (order matters: $B - A = \{5\}$); symmetric difference, in exactly one. Methods: union, intersection, difference, symmetric_difference; for arrays, np.union1d, np.intersect1d, np.setdiff1d. A and B are disjoint if $A \cap B = \emptyset$.

3 De Morgan: a complement swaps $\cup$ and $\cap$



The complement A^c is what the universe U holds outside A : with $U = \{1, \dots, 10\}$, $A^c = \{4, \dots, 10\}$, or $U - A$ in Python. De Morgan's laws: $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$. In probability, U is the sample space S .

4 The sample space lists every outcome



An experiment ends in exactly one outcome; the sample space S lists them all, with no overlaps. HT (first heads, then tails) and TH are different outcomes, so two coins give 4; two dice give $6 \times 6 = 36$ ordered pairs. `itertools.product("HT", repeat=2)` lists them.

5 An event is a subset of the sample space

event	faces	as a set	roll = 4
even	1 2 3 4 5 6	{2, 4, 6}	happened
at least 5	1 2 3 4 5 6	{5, 6}	no
a three	1 2 3 4 5 6	{3}	no
impossible	1 2 3 4 5 6	$\emptyset$	no
certain	1 2 3 4 5 6	S	happened

An outcome is one element, such as 4; an event is a set of outcomes, such as "even" = {2, 4, 6}. An event happens when the outcome lies in it. {3} is an elementary event, $\emptyset$ the impossible event, S the certain one. "A or B" is $A \cup B$, "A and B" is $A \cap B$, "not A" is A^c .

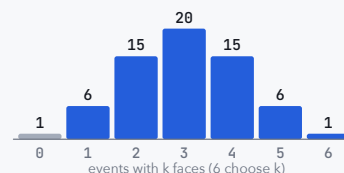
6 A die has 64 events: the power set

event {2, 4, 6}: each face in or out

1 2 3 4 5 6
0 1 0 1 0 1

$2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$ events

from {} to {1, ..., 6}



The event space is the set of all subsets of S , its power set. Each face is in or out, so a die has $2^6 = 64$ events, $\emptyset$ and S included; a coin has 4: $\emptyset$, {H}, {T} and {H, T}. `itertools.combinations(S, k)` for $k = 0, \dots, 6$ lists them all.

Probability laws: axioms and the rules that follow

A probability law gives every event a number. Three axioms fix the rules; the rest is derived from them.

7 Three axioms define a probability law

1 $P(A) \geq 0$
never negative

2 $P(S) = 1$
something happens

3 if $A \cap B = \emptyset$:
 $P(A \cup B) = P(A) + P(B)$

fair die:

1	2	3	4	5	6
1/6	1/6	1/6	1/6	1/6	1/6

 $P(\text{even}) = 1/2$

loaded (illustr.):

1	2	3	4	5	6
0.1	0.1	0.1	0.1	0.1	0.5

 $P(\text{even}) = 7/10$

A probability law P gives each event a number. Kolmogorov's axioms: $P(A) \geq 0$; $P(S) = 1$; probabilities of disjoint events add (for countably many too). On a finite S , give each outcome a non-negative mass, with masses summing to 1; an event gets the sum of its masses. They need not be equal: the loaded die (illustrative) has $P(\text{even}) = 7/10$.

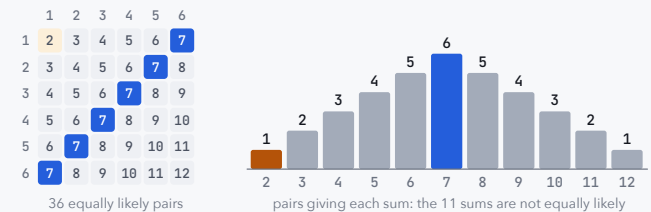
8 Rules that follow from the axioms

$A = \text{even}$
 $B = \text{at least 4}$

$P(A) = 3/6, P(B) = 3/6$
 $P(A \cap B) = 2/6$
 $P(A \cup B) = 3/6 + 3/6 - 2/6 = 4/6$
 $P(A \cap B^c) = 3/6 - 2/6 = 1/6$
 $P(A^c) = 1 - 3/6 = 3/6$
 $\{4, 6\} \subseteq A$, so $2/6 \leq 3/6$

A and A^c are disjoint and fill S , so $P(A^c) = 1 - P(A)$; hence $P(\emptyset) = 0$ and $P(A) \leq 1$. Split $A \cup B$ into the disjoint pieces $A \cap B^c$, $A \cap B$, $A^c \cap B$ and apply axiom 3: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, which removes the double count. If $A \subseteq B$, then $P(A) \leq P(B)$.

9 Equally likely outcomes: count and divide



With $|S|$ equally likely outcomes, $P(E) = |E| / |S|$. Two dice: 6 of 36 pairs sum to 7, so $1/6$; only $(1, 1)$ sums to 2, so $1/36$. Two coins: one of each = $\{HT, TH\}$, $2/4$. Trap: there are 11 sums, yet $P(\text{sum} = 7) \neq 1/11$; the pairs are equally likely, the sums are not. Bigger counts and $P(\text{at least one}) = 1 - P(\text{none})$ are on the counting sheet.

10 Mutually exclusive is not independent

1 **2** **3** **4** **5** **6**

$A = \{1, 2\}$ $C = \{5, 6\}$

$A \cap C = \emptyset, P(A \cap C) = 0$
 $P(A) P(C) = 1/3 \times 1/3 = 1/9$
exclusive, so not independent

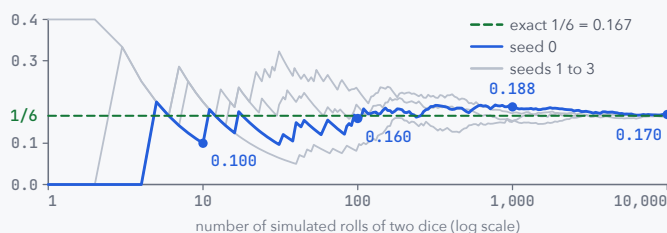
1 **2** **3** **4** **5** **6**

$A = \{1, 2\}$ $E = \text{even}$

$A \cap E = \{2\}, P(A \cap E) = 1/6$
 $P(A) P(E) = 1/3 \times 1/2 = 1/6$
independent, not exclusive

Exclusive (disjoint): $A \cap B = \emptyset$, so $P(A \cap B) = 0$ and probabilities add. Independent: $P(A \cap B) = P(A)P(B)$; one happening leaves the other's chance unchanged. Exclusive events with positive probabilities are dependent: A happening rules C out. More on independence in the conditional probability sheet.

11 Simulate to check an exact answer



Seed 0 gives 0.100 after 10 rolls, 0.160 after 100, 0.188 after 1,000 and 0.170 after 10,000. Other seeds take other paths to the same limit.

```
# exact: list S, keep E, divide
S = list(product(range(1, 7), repeat=2))
E = [s for s in S if sum(s) == 7]
Fraction(len(E), len(S)) # 1/6
# simulated: fraction of 10,000 rolls
rng = np.random.default_rng(0)
d = rng.integers(1, 7, size=(10_000, 2))
(d.sum(axis=1) == 7).mean() # 0.1697
```

The estimate wanders around $1/6$ and settles slowly. Enumerate when S is small enough to list; simulate when it is not, or to check a count.

12 Pause and think

How many events of a die contain the face 6? 32 : each of the other 5 faces is in or out, 2^5 .

Two coins: $P(\text{at least one head})$? $3/4$: only TT misses, so $1 - 1/4$.

$P(A) = 0.5, P(B) = 0.3, P(A \cup B) = 0.6. P(A \cap B^c)$? $P(A \cap B) = 0.2$, so $0.5 - 0.2 = 0.3$.

$\{1, 2, 3\} \wedge \{2, 3, 5\}$ gives? $\{1, 5\}$: elements in exactly one set.

Exclusive events with $P(A) = 0.7, P(B) = 0.5$? Impossible: $P(A \cup B)$ would be 1.2.

1,000 rolls gave 0.188, not 0.167. A bug? Maybe not: a typical miss at 1,000 rolls is about 0.012.

S lists the outcomes · an event is a subset · three axioms · complement and union rules · equally likely: $|E| / |S|$

Source: the course notebooks "Set" (Python sets, subsets, union, intersection, difference, complement, De Morgan) and "Probability" (sample spaces, the event space as a power set, a probability law by counting), nipunbatra.github.io/psdv-teaching. The other examples and all numbers are our own, computed by the sheet's figure script; the loaded die is illustrative.